THE CYCLE OF THE ZERO SCHEME OF A SECTION IS THE SUM OVER THE COMPONENTS OF THE CYCLES OF THE RESTRICTED SECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Additivity of divisors for a restricted family. Same chapter chow.tex, label lemma-additivity-divisors-restricted, invoked at proof line 4419. Supplies the final identity expressing a length over the local ring as the sum over its minimal primes of the local length times the length after quotienting by the prime, which is exactly the coefficient comparison the proof needs.

Source reference. Original lemma 02QG: sources/stacks-chow/pinned-original-source.tex, source lines 373–373. Navigation pointer for \ref{lemma-additivity-divisors-restricted} used inside the counted span (line 4420); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 4359); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let Y be locally of finite type over S. Let $\mathcal{L}$ be an invertible $\mathcal{O}_Y$-module. Let $s \in \Gamma(Y, \mathcal{L})$. Assume*

(1) $\dim_\delta(Y) \leq k + 1$,
(2) $\dim_\delta(Z(s)) \leq k$, *and*

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(3) *for every generic point ξ of an irreducible component of $Z(s)$ of δ-dimension k the multiplication by s induces an injection $\mathcal{O}_{Y,\xi} \to \mathcal{L}_\xi$.*

Write $[Y]_{k+1} = \sum n_i[Y_i]$ where $Y_i \subset Y$ are the irreducible components of Y of δ-dimension $k+1$. Set $s_i = s|_{Y_i} \in \Gamma(Y_i, \mathcal{L}|_{Y_i})$. Then

$$[Z(s)]_k = \sum n_i[Z(s_i)]_k \tag{0.1.1}$$

as k-cycles on Y.

Proof. Let $Z \subset Y$ be an integral closed subscheme of δ-dimension k. Let $\xi \in Z$ be its generic point. We want to compare the coefficient n of $[Z]$ in the expression $\sum n_i[Z(s_i)]_k$ with the coefficient m of $[Z]$ in the expression $[Z(s)]_k$. Choose a generator $s_\xi \in \mathcal{L}_\xi$. Write $A = \mathcal{O}_{Y,\xi}$, $L = \mathcal{L}_\xi$. Then $L = As_\xi$. Write $s = fs_\xi$ for some (unique) $f \in A$. Hypothesis (3) means that $f : A \to A$ is injective. Since $\dim_\delta(Y) \leq k+1$ and $\dim_\delta(Z) = k$ we have $\dim(A) = 0$ or 1. We have

$$m = \text{length}_A(A/(f))$$

which is finite in either case.

If $\dim(A) = 0$, then $f : A \to A$ being injective implies that $f \in A^*$. Hence in this case m is zero. Moreover, the condition $\dim(A) = 0$ means that ξ does not lie on any irreducible component of δ-dimension $k+1$, i.e., $n = 0$ as well.

Now, let $\dim(A) = 1$. Since A is a Noetherian local ring it has finitely many minimal primes $\mathfrak{q}_1, \ldots, \mathfrak{q}_t$. These correspond 1-1 with the Y_i passing through ξ'. Moreover $n_i = \text{length}_{A_{\mathfrak{q}_i}}(A_{\mathfrak{q}_i})$. Also, the multiplicity of $[Z]$ in $[Z(s_i)]_k$ is $\text{length}_A(A/(f, \mathfrak{q}_i))$. Hence the equation to prove in this case is

$$\text{length}_A(A/(f)) = \sum \text{length}_{A_{\mathfrak{q}_i}}(A_{\mathfrak{q}_i})\text{length}_A(A/(f, \mathfrak{q}_i))$$

which follows from Lemma 02QG. □